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Coimbatore – 641 032

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DEPARTMENT OF MECHANICAL ENGINEERING

PROJECT PHASE I - FIRST REVIEW

"PERFORMANCE ENHANCEMENT OF SOLAR WATER PUMP FOR IRRIGATION APPLICATION"

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ABSTRACT

- This Presentation is about the enhancement of Solar Water Pump for Irrigation application.
- In order to analyse the increased efficiency through Solar Energy.
- Agricultural applications suitable for Photovoltaic solutions are numerous.
- Solar Powered Irrigation system Can be adopted as a nature based solution to uplift water from the tank

INTRODUCTION

- In India, electrical and diesel-powered water pumping systems are widely used in agricultural applications.
- However, connecting the grid electricity with electrical water pumping systems in remote locations is more expensive.
- The idea of integrating the solar photovoltaic panels with water pumps has been proposed by many researchers around the world to meet the energy demand and to reduce the environmental impacts especially in agricultural sector.
- Earlier reported review studies confirmed that solar photovoltaic water pumping systems are having highest utilization in agriculture sector.
- The solar energy availability could fulfil the increasing power requirements specifically for agriculture water pumping applications in a decentralized, efficient and sustainable manner in the form of solar photovoltaic energy conversion systems

OBJECTIVES

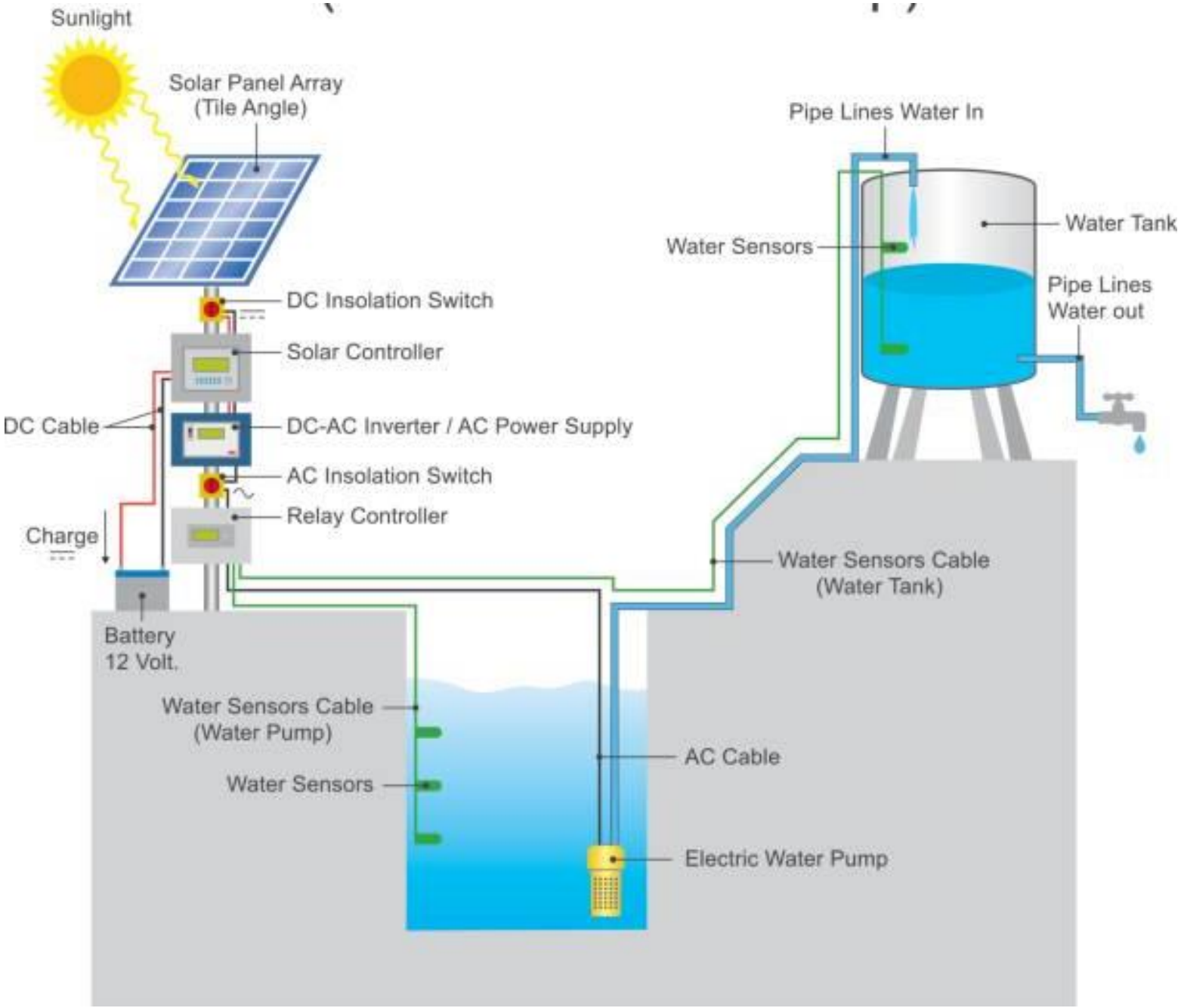
- To investigate the energy performance of the solar photovoltaic water pumping systems under the influence of air cooling and water cooling at the climatic conditions in Coimbatore.
- To develop a mathematical model for predicting the energy performance under different climatic conditions.
- To evaluate the technical, economical and environmental benefits of solar photovoltaic water pumping applications.
- To evaluate its feasibility for irrigation applications.

METHODOLOGY

- Review of studies related to the photovoltaic water pumping systems.
- Problem was identified based on the literature review.
- Experimental test facility was developed with necessary instrumentation.
- Experiments were carried out under different climatic conditions.
- Development of a mathematical model for performance prediction.
- Performance evaluation of water pumping system under the influence of panel cooling.
- Economical evaluation for the period of twenty years.
- Environmental benefits are assessed for the period of twenty years.

WORKING

- When solar energy (Light energy) falls on the photovoltaic panels it gets converted into electrical energy in the form of DC. Now this DC can either drive a DC motor coupled with a pump or can be inverted through an inverter so that it can drive an AC motor.
- The coupled pump will lift water from low to high. It may be noted that solar power is available only during the day time with clear sky therefore if we need irrigation or power at night time we have to have a storage device that could be either a battery or a pumped storage arrangement.
- In the pumped storage arrangement, we lift the water and store it at a high elevation during the day and in the night when solar power is not available, we get a gravity flow for irrigation.


Sunlight
Solar Panel Array
(Tile Angle)
Pipe Lines Water In
Water Tank
Water Sensors
Pipe Lines
Water out
DC Insolation Switch
Solar Controller
DC Cable
DC-AC Inverter / AC Power Supply
AC Insolation Switch
Relay Controller
Charge
Water Sensors Cable
(Water Tank)
Battery
12 Volt.
Water Sensors Cable
(Water Pump)
AC Cable
Water Sensors
Electric Water Pump

COMPONENTS

The following components are used:

- Solar panel
- Battery
- pump
- Control unit

Solar panel

- Photovoltaic modules consist of a large number of solar cells and use light energy (photons) from the Sun to generate electricity through the photovoltaic effect. Most modules use wafer-based crystalline silicon cells or thin-film cells

- The structural (load carrying) member of a module can be either the top layer or the back layer. Cells must be protected from mechanical damage and moisture. Most modules are rigid, but semi-flexible ones based on thin-film cells are also available.
- The cells are usually connected electrically in series, one to another to the desired voltage, and then in parallel to increase current. The power (in watts) of the module is the mathematical product of the voltage (in volts) and the current (in amperes) of the module. The manufacturing
- The manufacturing specifications on solar panels are obtained under standard conditions, which is not the real operating condition the solar panels are exposed to on the installation site.

BATTERY

(Lead Acid battery)

- A lithium-ion or Li-ion battery is a type of rechargeable battery which uses the reversible reduction of lithium ions to store energy. It is the predominant battery type used in portable consumer electronics and electric vehicles. It also sees significant use for grid-scale energy storage
- As they are inexpensive compared to newer technologies, lead–acid batteries are widely used even when surge current is not important and other designs could provide higher energy densities.

- The lead–acid cell can be demonstrated using sheet lead plates for the two electrodes. However, such a construction produces only around one ampere for roughly postcard-sized plates, and for only a few minutes. One unfortunate disadvantage of lead-acid batteries is that the chemical reaction described above can never be halted completely.
- In other words, these batteries will continue to discharge even when they're not in use.
- A lead-acid battery is a rechargeable battery that uses lead and sulphuric acid to function. The lead is submerged into the sulphuric acid to allow a controlled chemical reaction.
- This chemical reaction is what causes the battery to produce electricity. Then, this reaction is reversed to recharge the battery.

PUMP

- Utilizing renewable sources of power like solar energy will get a lot of impetus in days to come.
- Irrigation through pumping is a must for our agriculture growth.
- By using solar energy for pumping we can grow food without harming our environment. It may be added that India being a tropical country has an abundance of sunshine, which, can cost-effectively be used for generating power to meet our pumping needs.
- Solar pumping can be done either thru direct heat pumps (utilizing heat energy of the sun) or through electrical power generated solar energy.

ADVANTAGES

- Solar pumps have no fuel cost as they run on renewable clean energy. Thus, they have a very low running cost. The spiraling grid cost has drastically reduced the payback period of these pumps.
- Solar pumps can be installed locally without grid connectivity. Getting a grid connection is difficult and leads to very heavy transmission losses and theft of power.
- The environmental impact of utilizing solar energy is enormous.
- Solar pumps can be installed and maintained easily as compared to pumps using fossil fuels. (especially surface mounted pumps)
- These pumps can be shifted easily from one place to another.

APPLICATIONS

- Solar pumps are mostly used for irrigation purposes but can be put to other application as well
- Direct irrigation solar pumps can be used pumped-storage which can be used during peak hours or when solar ener
- Ideally suited for all our pumping needs including drinking water where grid supply cannot reach due to difficult terrain
- Used in water treatment plants.
- Used as booster pumps for transporting liquid over long distances.

CONCLUSION

- The analytical model predicts the energy performance with ± 5% deviations.
- ANOVA results showed that solar intensity and pumping head are the most influencing parameters affecting the performance.
- The ambient wind velocity and ambient temperature has minor influence.
- The relative humidity has negligible influence on performance.
- The economic assessment confirms that, the investment in photovoltaic water pumping system is economically feasible.
- The environment impact assessment confirmed that, the solar photovoltaic water pumping system has significant reduction in environmental impacts due to the reduction in carbon-dioxide emissions.

REFERENCE

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THANK YOU